

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	CO2	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	ADT array, Operations on arrays, Searching Algorithms, Linear Search, Binary Search, Suffix Arrays & LCP Arrays ADT, Linked List, Stack, Queue, Applications.		
Student Name:	Peram.Mahitha	Reg. No.:	192524136
Section / Batch:	2025	Date:	28-07-2026

Code of Conduct

I Peram.Mahitha(192524136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____P.Mahitha_____

Instructions

- This test contains 3 Analytical Problem Solving questions. Total: 100 Marks.
- When a student answer using a analytical problem, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are not permitted in the class. Take it as test and submit in the class.

Section / Topic	Focus	Q Nos	Marks
Stack Data Structures	Stack – top movements	1	30
Queue Data Structures	Circular Queue	2	40
Application of Stack	Infix to post-fix	3	40
GRAND TOTAL:		100 Marks	

Annexure A – Analytical Problem Solving

1) Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of the data structure and its index position. With a neat sketch, explain in detail.

2) Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

3) Convert the following infix expression into a post fix expression.

$$B - (C/D + (E \% F * G) / H) * J.$$

Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 /*

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

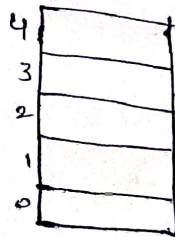
28-07-2024

Date: 28/07/2026.
Peram. Mahitha
192524136

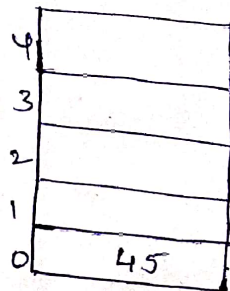
Analytical Problem Solving - CO2 AT1 (Test)

- ① Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), pop(), push(44), push(33), pop(), pop(), pop(). Print the peak element of the data structure and its index position. With a neat sketch explained in detail.

Given an array size of 5

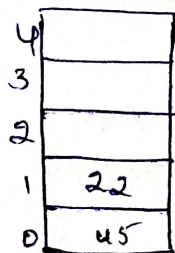


(i) push(45)



$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= -1 + 1 \\ &= 0 \end{aligned}$$

(ii) push(22)



$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

iii) push(11)

4	
3	
2	11
1	22
0	45

$$\text{top} = \text{top} + 1$$

$$= 1 + 1$$

$$\text{top} = 2$$

iv) push(77)

4	
3	77
2	11
1	22
0	45

$$\text{top} = \text{top} + 1$$

$$= 2 + 1$$

$$\text{top} = 3$$

v) pop()

4	
3	
2	11
1	22
0	45

$$\text{top} = \text{top} - 1$$

$$= 3 - 1$$

$$= 2$$

(vi) push(44)

4	
3	44
2	11
1	22
0	45

$$\text{top} = \text{top} + 1$$

$$= 2 + 1$$

$$\text{top} = 3$$

(vii) push(33)

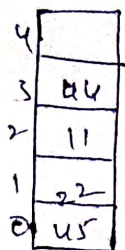
4	33
3	44
2	11
1	22
0	45

$$\text{top} = \text{top} + 1$$

$$= 3 + 1$$

$$\text{top} = 4$$

pop()

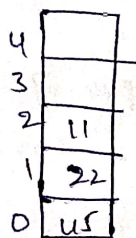


$$\text{top} = \text{top} - 1$$

$$= 3 - 1$$

$$\text{top} = 3$$

(ix) pop()

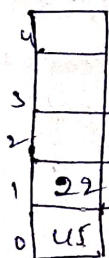


$$\text{top} = \text{top} - 1$$

$$= 3 - 1$$

$$\text{top} = 2$$

(x) pop()



$$\text{top} = \text{top} - 1$$

$$= 2 - 1$$

$$\text{top} = 1$$

Peak element = 22

22 is at position 1.

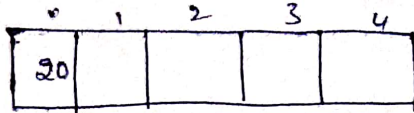
② Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(70), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

size = 5

front = Rear = -1

(i)

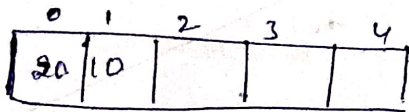
Enqueue (20)



FR

(ii)

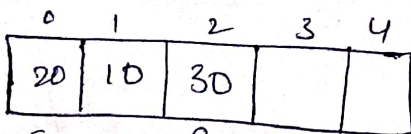
Enqueue (10)



F R

(iii)

Enqueue (30)

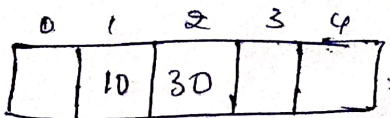


F

R

(iv)

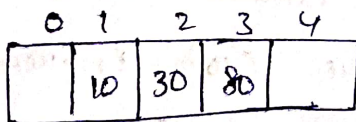
Dequeue ()



F R

(v)

Enqueue (80)

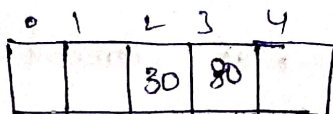


F

R

(vi)

Dequeue ()



F

R

f = 0, R = 0

Que[Rear] = 20

F = 0, R = 1

Queue[Rear] = 10

Rear = Rear + 1

= 0 + 1

Rear = 1

f =

Queue[Rear] = 30

Rear = Rear + 1

= 1 + 1

Rear = 2, f = 0

Queue[Rear]

Front = Front + 1

= 0 + 1

front = 1, R = 2

front = 1, Rear = Rear + 1

= 2 + 1

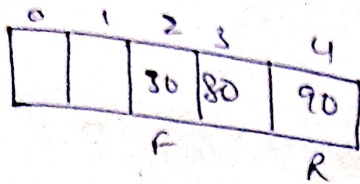
= 3

front = front + 1, Rear = 3

= 1 + 1

= 2

enqueue (90)



$$\text{front} = 2$$

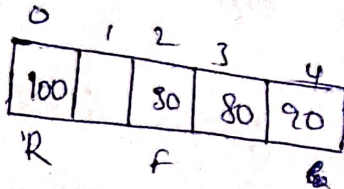
$$\text{Rear} = \text{Rear} + 1$$

$$= 3 + 1$$

$$\text{Rear} = 4$$

(vii)

enqueue (100)



$$\text{front} = 2$$

$$\text{Rear} = \text{Rear} + 1$$

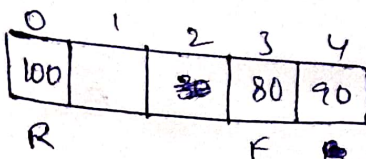
$$= -1 + 1$$

$$= 0$$

Queue is full

(ix)

Dequeue()



$$\text{front} = \text{front} + 1$$

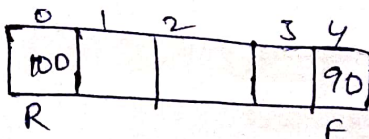
$$\text{Rear} = 0$$

$$= 2 + 1$$

$$= 3$$

(x)

Dequeue()



$$\text{front} = \text{front} + 1$$

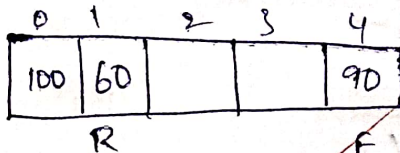
$$\text{Rear} = 0$$

$$= 3 + 1$$

$$= 4$$

(xi)

enqueue (60)



$$\text{front} = 4$$

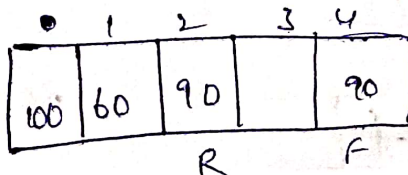
$$\text{Rear} = \text{Rear} + 1$$

$$= 0 + 1$$

$$= 1$$

(xii)

enqueue (90)



$$\text{front} = 4$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 1 + 1$$

$$= 2$$

front element 90 at position '4'
Rear element 90 at position '2'

③

Convert the following infix expression into a postfix expression.

$$B - (C/D + (E \% F * G) / H)^A J$$

Also evaluate the given postfix expressions.

(a) $a \ 3 \ 84 \ / \ +$ b) $6 \ 2 \ + \ 6 \ 3 \ / \ *$

Scanning character

stack

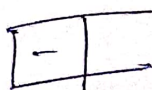
postfix expression

B



B

-



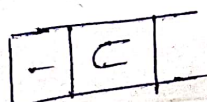
B

(



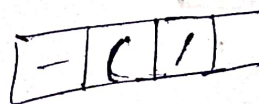
B

C



BC

/



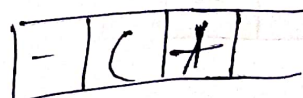
BC

D



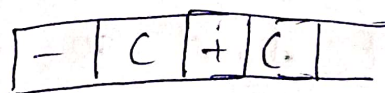
BCD

+



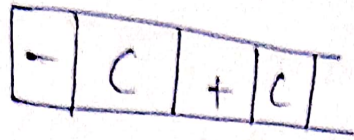
BCD/

(



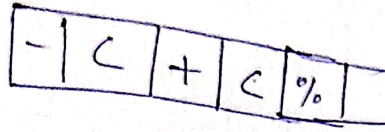
BCD/

f
%



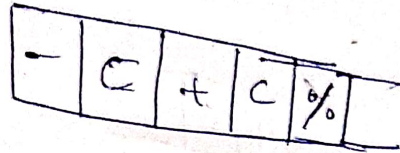
BCD/E

F



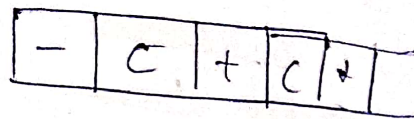
BCD/E

*



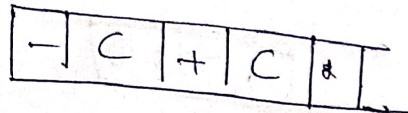
BCD/EF

G



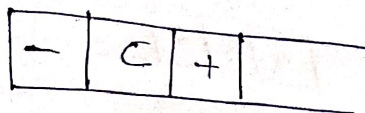
BCD/EF%

)



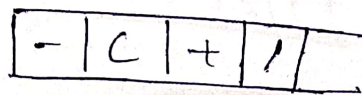
BCD/EF%G

/



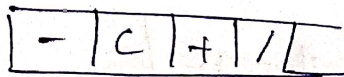
BCD/EF%G⁰

+



BCD/EF%G¹

)



BCD/EF%G⁰H

*



BCD/EF%G¹H

J



error
thin

BCD/EF%G¹H/+

OVER



BCD/EF%G¹H/+J



BCD/EF%G¹H/+J¹

The output is

BCD/EF%G¹H/+J¹

(a) 9 3 8 4 / * +

Scenary char

stack

operation

9

9

3

9 3

8

9 3 8

4

9 3 8 4

/

9 3 2

$8/4 = 2$

*

9 6

$3 \times 2 = 6$

+

15

$9 + 6 = 15$

output = 15

(b)

6 2 + 6 3 / *

stack

operation

6

6

2

6 2

+

8

$6 + 2 = 8$

6

8 6

3

8 6 3

/

8 2

$6/3 = 2$

*

16

$8 \times 2 = 16$

output = 16

